

|                  |                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                 |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
|                  | $\gamma = 0.50 \text{ s}^{-1}$                                                                                                                                                                                                                                                                                                                                                   |                                                                                                 |
| <b>8(a)(i)</b>   | $A = 94, Z = 35$                                                                                                                                                                                                                                                                                                                                                                 | [1] both correct                                                                                |
| <b>8(a)(ii)</b>  | <p>Mass defect of U-235, <math>m</math></p> $= (143)(1.00866) + (92)(1.00728) - (235.044)$ $= 1.864 \text{ u}$ <p>Binding energy of U-235 <math>= mc^2</math></p> $= (1.864)(1.66 \times 10^{-27})(3.00 \times 10^8)^2$ $= 2.785 \times 10^{-10} \text{ J}$ $= 1741 \text{ MeV}$ <p><b>OR</b></p> <p>Binding energy of U-235 <math>= 1.864 (934)</math></p> $= 1741 \text{ MeV}$ | <p>[1] value for <math>m</math></p> <p>[1] sub</p> <p>[1] ans</p> <p>[1] sub</p> <p>[1] ans</p> |
| <b>8(a)(iii)</b> | Neutron(s) is/are produced                                                                                                                                                                                                                                                                                                                                                       | [1] ans                                                                                         |
| <b>8(b)(i)</b>   | <u>High energy</u> electron                                                                                                                                                                                                                                                                                                                                                      | [1] ans                                                                                         |

